FIN101 Lecture 3: The Time Value of Money

SAMPLE CONTENT: written for the Gen AI demo. Not real course material.

1. Why money has a time value

A baht today is worth more than a baht received in the future. Money available now can be invested to earn interest, and future payments carry risk and lose purchasing power to inflation. The time value of money (TVM) is the idea that we must adjust for timing before comparing cash flows that occur on different dates.

Two operations do this adjustment. Compounding moves a value forward in time to find its future value. Discounting moves a value back in time to find its present value. The rate used is called the interest rate when compounding and the discount rate when discounting.

2. Future value and compound interest

With simple interest, interest is earned only on the original principal. With compound interest, interest is also earned on previously earned interest. The future value of a single amount is FV = PV x (1 + r)^n, where PV is the present value, r is the interest rate per period and n is the number of periods.

Worked example: 10,000 baht invested at 5% per year for 3 years grows to 10,000 x 1.05^3 = 11,576.25 baht. With simple interest it would grow to only 11,500 baht. The extra 76.25 baht is interest earned on interest.

When interest is compounded m times per year, use the periodic rate r/m and the number of periods n x m. More frequent compounding gives a higher future value. The effective annual rate is EAR = (1 + r/m)^m - 1.

3. Present value and discounting

The present value of a future amount is PV = FV / (1 + r)^n. A higher discount rate or a longer waiting time makes the present value smaller. For example, 20,000 baht to be received in 4 years, discounted at 6% per year, is worth 20,000 / 1.06^4 = 15,841.87 baht today.

Common mistake: using the annual rate with monthly periods. If cash flows are monthly, convert both the rate and the number of periods to months before applying the formula.

Annuities are equal payments made every period. The present value of an ordinary annuity is PV = PMT x [1 - (1 + r)^-n] / r. A loan repayment schedule is an annuity viewed from the lender's side.

4. FIN101 course policies (invented for the demo)

Problem sets are due every Friday at 17:00. A late problem set loses 15% of its marks for each day it is late, and problem sets more than 3 days late receive zero marks.

The midterm exam covers Lectures 1 to 4 and allows one A4 formula sheet written by hand on both sides. Financial calculators are allowed; phones are not.

Office hours are on Wednesdays from 13:00 to 15:00 in room FIN-204, or online by appointment.